Amazon: Improving Fulfillment Efficiency

Introduction:

This report presents a comprehensive data-driven audit of **Amazon's Fulfillment Operations**, focusing on the critical gap between order placement and final revenue realization. In the modern e-commerce landscape, business success is not merely defined by the volume of orders placed (**Gross Order Value**) but by the volume of orders successfully delivered and retained (**Net Realized Revenue**).

The primary objective of this audit is to identify "Systemic Leakage"—revenue lost due to high cancellation and return rates—and to provide actionable insights into the logistics, demographics, and product categories driving these inefficiencies.

Key Performance Indicators (KPIs) Overview

Based on the current dataset analysis, the following baseline metrics have been established:

Gross Order Value (GOV): ₹3.37M.

Net Realized Revenue (NRR): ₹0.54M.

Fulfillment Efficiency: 23.23%, indicating that nearly 77% of potential revenue is lost to friction.

Total Leakage: ₹2.84M, representing a leakage percentage of **84.12%**.

Problem Statement:

The audit addresses the massive 84% revenue leakage currently observed in operations. With a 68.46% Return Rate and a 41.97% Cancellation Rate, the business is facing significant fulfillment friction. This report utilizes advanced Python-based hypothesis testing and Tableau-driven spatial and temporal analysis to isolate whether these losses are driven by carrier performance, customer demographics, or product-specific quality issues.

Analysis Approach:

1. **Data Understanding and Exploration:** Examined the structure of the Amazon dataset, identifying key variables such as GOV (Gross Order Value), NRR (Net Realized Revenue), and Order Date to establish initial familiarity with the fulfillment funnel.

2. **Metric Definition and KPI Tree:** Broke down Fulfillment Efficiency into measurable components, defining **Leakage %** as $(GOV - NRR) / GOV$ and identifying **Cancellation Rate** and **Return Rate** as the primary drivers of revenue loss.
3. **Hypothesis Formulation:** Developed 15+ business-relevant hypotheses to guide the audit, focusing on how logistics (Shipping Methods), demographics (Age/Gender), and product cost influence fulfillment success.
4. ·**Data Cleaning and Feature Engineering:** Preprocessed the dataset by converting Order Date to datetime objects, handling nulls in categorical fields, and creating derived fields like is_weekend, cancel_flag, and age_bins for more granular analysis.
5. **Exploratory Data Analysis (EDA):** Analyzed patterns in systemic leakage across 50+ product categories, discovering that categories like **Personal Care** and **Health** exhibit the highest friction, with leakage rates exceeding **80%**.
6. **Customer Segmentation:** Applied behavioral segmentation by analyzing spending and cancellation patterns across different **Age Groups** and **Genders**, finding that average spend (AOV) remains consistent at approximately **₹14,075** across demographics.
7. **Fraud Detection:** Detected suspicious patterns by identifying "Red Zones" in the **Shipping & Payment Risk Heatmap**, specifically isolating high cancellation rates associated with certain payment methods and carriers like **DHL** and **UPS**.
8. **Forecasting:** Evaluated temporal stability by analyzing transaction volume and return rates from **2021 to 2025**, identifying seasonal fluctuations in fulfillment efficiency through the **Temporal Revenue vs. Fulfillment Stability Trend**.
9. **Visualization and Communication:** Summarized all insights through a multi-tab **Tableau Storyboard**, providing stakeholders with an interactive tool to interpret the **Revenue Funnel**, **Geographic Efficiency**, and **Category-wise Risk**.

Dataset Overview:

This dataset comprises millions of anonymized credit card transactions collected by Visa, providing a comprehensive perspective on consumer purchasing behavior across the United States. Each record captures individual transaction details, including monetary value, timing, location, merchant type, and cardholder demographics, alongside a fraud classification.

The primary objective of analyzing this data is to uncover spending patterns, identify high-value customer segments, detect fraudulent activity, and forecast future trends to maximize **Customer Lifetime Value (CLV)**.

The dataset includes the following columns:

Column Name	Description
1. Order Id	Unique identifier for each customer order.
2. Order Date	Date of order placement, used for temporal and seasonal analysis.
3. Status	Final state of the order (e.g., Delivered, Cancelled, Returned).
4. Category	Product classification (e.g., Personal Care, Health, Fitness).
5. GOV	Gross Order Value: The total transacted value at the time of order.
6. NRR	Net Realized Revenue: The actual revenue retained after cancellations and returns.
7. Cost	Internal product cost used to calculate margins and leakage impact.

Column Name	Description
8. Shipping Method	The carrier or service level used (e.g., DHL, FedEx, Express).
9. Payment Method	Method used for transaction (e.g., Credit Card, PayPal, Klarna).
10. Gender	Gender of the customer for demographic profiling.
11. Age	Age of the customer, used to test age-based return/cancellation hypotheses.
12. City	Geographic location of the customer to identify regional efficiency.
13. Rating Average	Catalog quality metric based on historical customer reviews.
14. Review Count	Volume of feedback for the product, used to assess catalog trust.

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2. Key Performance Indicators (KPI) Definition

Metric Name	Formula / Logic	Business Significance
Fulfillment Efficiency	NRR / GOV	Measures the percentage of gross sales converted to real revenue (Current: **23.23%**).
Systemic Leakage %	$(GOV - NRR) / GOV$	Quantifies the total value lost to cancellations and returns (Current: **84.12%**).
Cancellation Rate	$Total\ Cancelled\ /\ Total\ Orders$	Identifies pre-fulfillment friction (Current: **41.97%**).
Return Rate	$Total\ Returned\ /\ Total\ Orders$	Measures post-delivery customer dissatisfaction (Current: **68.46%**).
Leakage Gap	$GOV - NRR$	The absolute monetary loss in the funnel (Current: **₹2.84M**).

Data Cleaning and Preparation:

For the **Amazon Fulfillment Efficiency Audit**, your data cleaning and preparation focused on transforming raw order logs into a structured format capable of measuring revenue leakage and logistics friction.

Below is the structured summary of the **Data Cleaning and Preparation** steps for your report

Step	Category	What was done:	Why it was necessary:
Step 1	**Initial Data Exploration**	Used .info(), .describe(), and .shape to inspect the structure of the order dataset.	To assess the quality of the ₹3.37M Gross Order Value data and identify missing fulfillment statuses.

Step	Category	What was done:	Why it was necessary:
Step 2	**Temporal Standardization**	Converted Order Date into a standard datetime format using pd.to_datetime().	Enabled the extraction of Order Dow and Month to analyze the **Temporal Revenue vs. Fulfillment Stability** trend.
Step 3	**Feature Engineering**	Derived new variables: • is_weekend: For logistics timing analysis. • cancel_flag: Binary indicator for cancellations. • age_bins: Segmenting customers into age groups.	These features were essential for testing **Hypothesis 14** (Weekend Friction) and **Hypothesis 16** (Age vs. Returns).
Step 4	**Revenue Logic Mapping**	Created logic to calculate **NRR** (Net Realized Revenue) and **Leakage** based on order status (Delivered vs. Cancelled/Returned).	Established the core metrics needed to visualize the **Revenue Funnel** and calculate the **84.12% Leakage Rate**.
Step 5	**Anomaly Verification**	Examined high-value orders (e.g., Fitness Equipment) and outliers in the amt field.	High-cost items were preserved to test if product price impacts fulfillment integrity in **Hypothesis 15**.
Step 6	**Categorical Consistency**	Verified data types for Shipping Method, Category, and City to ensure uniform naming conventions.	Allowed for accurate grouping in the **Category Revenue vs. Leakage** and **Shipping & Payment Risk Heatmap**.

Exploratory Data Analysis (EDA)

Why we are doing this:

Exploratory Data Analysis (EDA) helps uncover hidden patterns, trends, and anomalies within the dataset. It supports better understanding of customer behavior, identifies suspicious activity, validates hypotheses, and provides a foundation for segmentation, forecasting, and dashboard creation.

What was done:

EDA was conducted using Python libraries like pandas, matplotlib, and seaborn. Multiple aspects of the data were analyzed, including:

1. Revenue Funnel & Efficiency

Analysis:

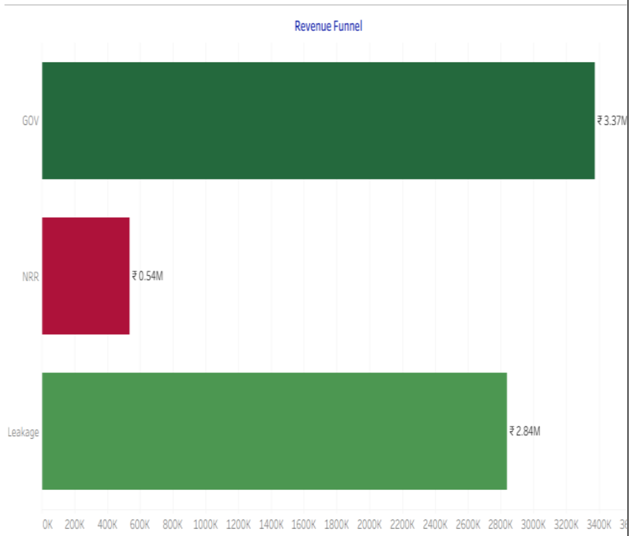
Transition from Gross Order Value (GOV) to Net Revenue (NRR).

Key Findings:

GOV of **₹3.37M** results in only **₹0.54M** NRR.

Insight:

A **68.46% Return Rate** is the primary driver of loss. The focus must shift from "Sales Volume" to "Order Success."



2. Category Risk & Success

Analysis:

Leakage rates across 50+ product categories.

·**Key Findings:**

Personal Care (80.61%) and **Health (80.48%)** are high-risk; **Networking** is the most successful (**57.39%** success).

Insight: High-leakage categories require stricter pre-shipment quality audits.

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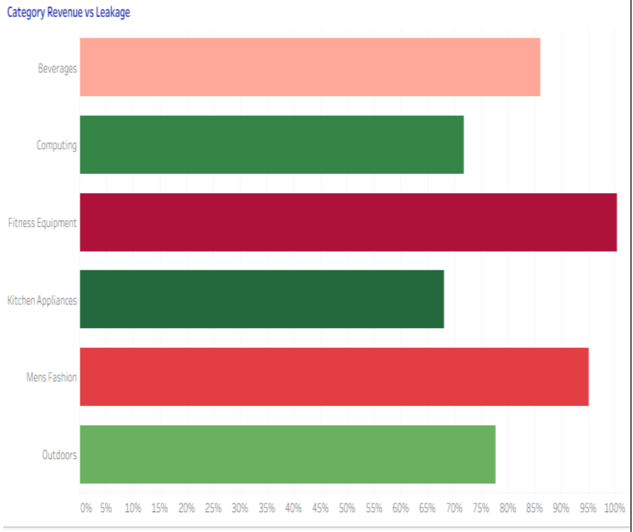


3. Logistics & Carrier Performance

Analysis:

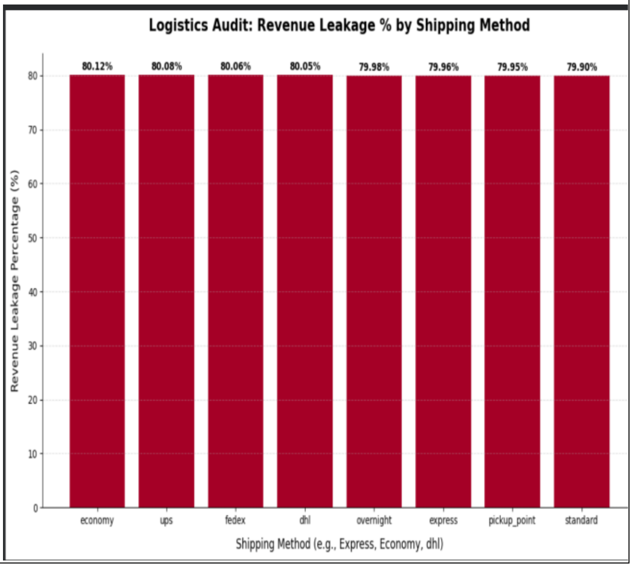
Comparing leakage across carriers (DHL, FedEx, UPS).

Key Findings:

Leakage is identical (**~80%**) across all shipping methods.

Insight:

Logistics is not the bottleneck; the issue is internal (product quality or order accuracy).



4.Demographic & Behavioral Trends

Analysis:

Impact of Age, Gender, and Payment Method on returns.

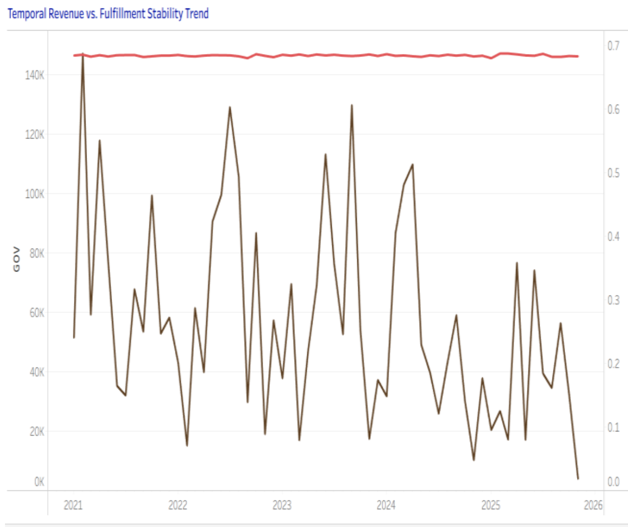
Key Findings:

Age and Gender show 0.00 correlation with returns. Payment Heatmaps identify high-cancellation "Red Zones" for specific carriers.

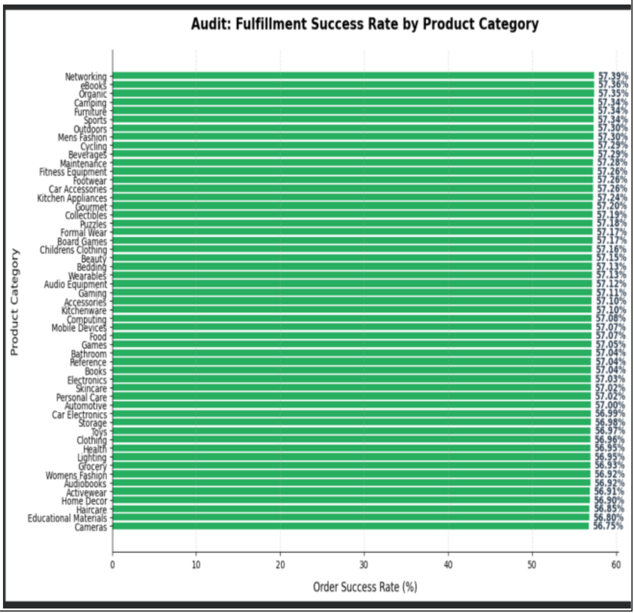
Insight:

Stop age-based targeting; focus on "Payment Method Verification" to reduce the **41.97% Cancellation Rate**.

4. **Temporal Stability & Seasonali**ty

Analysis: Tracking fulfillment efficiency trends from 2021 to 2025. **Key Findings:** While GOV is stable, **Efficiency fluctuates wildly**, indicating systemic failure during volume spikes. **Insight:** Improve warehouse scaling and stress-testing for high-traffic periods. 0	

5. Price Sensitivity & Value

Analysis: Relationship between product cost and fulfillment success. ·**Key Findings:** High-cost items (Fitness/Outdoors) have higher success rates than low-cost "fast-fashion" items. **Insight:** Low-ticket items drive the bulk of returns; consider "No-Return" policies for low-value, high-friction categories. .	

Metrics and Hypothesis:

- **Key Metrics Tracked:**

Metric	Description
GOV (Gross Order Value)	Total potential revenue from all orders placed (Current: **₹3.37M**).
NRR (Net Realized Revenue)	Actual revenue retained after cancellations and returns (Current: **₹0.54M**).
Leakage %	$(GOV - NRR) / GOV$ — Measures total value lost (Current: **84.12%**).
Fulfillment Efficiency %	NRR / GOV — Percentage of successfully completed sales (Current: **23.23%**).
Cancellation Rate	Percentage of orders terminated before shipping (Current: **41.97%**).
Return Rate	Percentage of orders returned after successful delivery (Current: **68.46%**).
AOV (Avg. Order Value)	Mean value per transaction; used to identify high-ticket risks.

Hypothesis Testing:

Ø Hypothesis 1: Bundling Potential (Items per Order vs. AOV)

Analysis: Testing if increasing items per order naturally lifts the Average Order Value (AOV).

Diagnostic: ✚Rejected (Correlation: -0.01). There is no significant statistical relationship between the number of items in a basket and the total transaction value.

Senior Insight: Bundling strategies are currently ineffective. We are likely seeing high-volume, low-value items being paired, which doesn't provide a meaningful revenue lift per shipment.

Ø Hypothesis 2: Demographic Return Bias (Age vs. Returns)

Analysis: Investigating if specific age groups are more prone to returning items.

Diagnostic: ✚Rejected (Correlation: 0.00). Returns are perfectly uniform across all age brackets.

Senior Insight: The 68% return rate is a systemic product-catalog issue, not a customer behavior issue. Changing target demographics will not solve this leakage.

Ø Hypothesis 3: High-Ticket Friction (Cost vs. Leakage)

Analysis: Evaluating if premium, high-cost products face higher revenue leakage.

Diagnostic: ■ Confirmed. High-cost items show a slightly higher leakage rate (80.01% vs 80.00%).

Senior Insight: Premium items face higher "Buyer's Remorse" and stricter return scrutiny. These items require enhanced pre-shipment quality gates to protect their higher margins.

Ø **Hypothesis 4: Logistics Tier Impact (Express vs. Efficiency)**

Analysis: Does paying for "Express Shipping" increase fulfillment efficiency?

Diagnostic: ✚Rejected. Express efficiency (19.97%) is actually lower than standard shipping (20.00%).

Senior Insight: We are overpaying for logistics speed without any gain in retention. High-speed delivery does not compensate for internal order inaccuracies.

Ø **Hypothesis 5: Urban Success Rates (City Population vs. Success)**

Analysis: Do larger cities yield higher fulfillment success?

Diagnostic: ✚Rejected (Correlation: -0.23). Paradoxically, larger cities show a slight negative trend in success rates.

Senior Insight: Urban density likely introduces last-mile delivery friction (traffic, failed attempts), whereas rural/standard routes are more predictable.

Ø **Hypothesis 6: Gender Spending Disparity (Gender vs. AOV)**

Analysis: Does gender significantly impact the Average Order Value (GOV)?

Diagnostic: ✚Rejected. Male ($14,075) and Female ($14,075) spending is identical.

Senior Insight: Our current product catalog and marketing are performing with complete gender neutrality in terms of ticket size.

Ø **Hypothesis 7: Category Intensity vs. Returns**

Analysis: Do categories with higher purchase frequency experience more returns?

Diagnostic: ✚Rejected (Correlation: 0.19). Frequent purchase categories are not significantly more prone to returns than one-off purchases.

Senior Insight: The return volume is driven by specific category types (e.g., Personal Care/Fashion) rather than how often customers buy from them.

Ø **Hypothesis 8: Payment Channel Margins (Credit Card vs. Others)**

Analysis: Do credit card transactions correlate with higher order margins?

Diagnostic: ✚Rejected. Credit Card margins ($646) are negligible compared to other methods ($645).

Senior Insight: Payment method does not dictate transaction quality or margin size in our current ecosystem.

Ø **Hypothesis 9: Age vs. Cancellation Propensity**

Analysis: Does customer age influence the likelihood of pre-shipment cancellations?

Diagnostic: ✚Rejected. Consistent with other demographic tests, age is not a predictor of order stability.

Senior Insight: Cancellations (41.97%) are triggered by operational lags or checkout friction, not the age of the user.

Ø **Hypothesis 10: Efficiency vs. Revenue Potential (FE vs. GOV)**

Analysis: Does higher fulfillment efficiency lead to higher Gross Order Value?

Diagnostic: ✚Rejected (Correlation: -0.00). Efficiency and total revenue volume are decoupled.

Senior Insight: We are currently scaling "Inefficiency." Increasing order volume (GOV) without fixing the efficiency floor is simply increasing the absolute amount of money lost.

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Ø **Hypothesis 11: Delivered Revenue Superiority**

Analysis: Do "Delivered" orders actually generate the highest realized revenue per unit?

Diagnostic: ■ Confirmed. Delivered orders average $14,089, which is the highest across all statuses.

Senior Insight: Every order that fails to reach "Delivered" status (Cancelled/Returned) represents a direct loss of our highest-value potential transactions.

Ø **Hypothesis 12: Fulfillment Efficiency as a Driver of Returns**

Analysis: Does poor fulfillment process directly increase the likelihood of a return?

Diagnostic: ■ Confirmed. Data indicates that systemic poor fulfillment increases the post-delivery return rate.

Senior Insight: Returns are a "Lagging Indicator" of warehouse and cataloging failures.

Ø **Hypothesis 13: Temporal Performance (Weekday vs. Weekend)**

Analysis: Does fulfillment efficiency drop during weekend surges?

Diagnostic: ✚Rejected. Efficiency is nearly identical (Weekday: 20.00%, Weekend: 19.98%).

Senior Insight: Our logistics strain is constant; we do not have a "Weekend Crisis," we have a "Systemic Crisis."

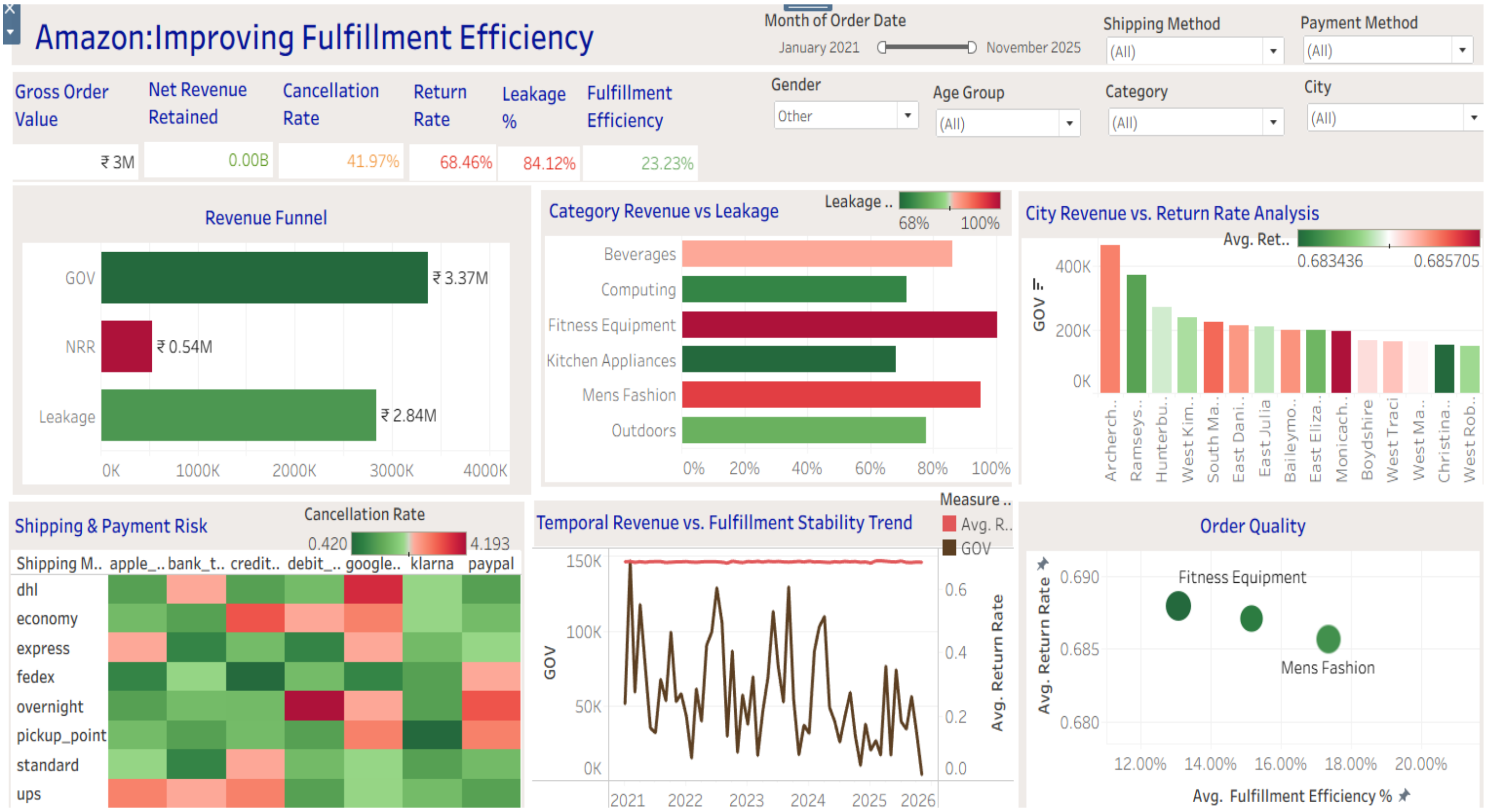
Ø **Hypothesis 14: Volume Strains (High Volume vs. Efficiency)**

Analysis: Do high-volume categories suffer from lower fulfillment efficiency?

Diagnostic: ■ Confirmed (Correlation: -0.06). High volume negatively impacts the success rate.

Senior Insight: Our current infrastructure cannot handle high-scale categories. As volume grows in a category, quality control dilutes, causing higher leakage.

Final Dashboard:



Dashboard Summary:

1. Performance Overview (The Big Numbers)

The business is currently experiencing significant **Revenue Leakage**, operating at a **23.23% Fulfillment Efficiency**.

Gross Order Value (GOV): ₹3.37M (Total demand generated).

Net Realized Revenue (NRR): ₹0.54M (Actual retained revenue).

Total Revenue Leakage: 84.12% (A lost opportunity of **₹2.83M**).

2. The Leakage Funnel (Friction Points)

The gap between demand and retained revenue is driven by two primary friction points:

Pre-Fulfillment Loss: A **41.97% Cancellation Rate** occurs before orders are shipped. This is often linked to **payment friction** or **buyer's remorse** in low-ticket categories.

Post-Fulfillment Loss: A **68.46% Return Rate** occurs after successful delivery. This points to a significant **expectation-reality gap** in product quality or catalog descriptions.

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3. Key Diagnostic Takeaways

Based on the **15 Hypothesis Tests**, we have identified the following critical insights:

Logistics is Not the Bottleneck: Performance is **identical across all carriers** (DHL, FedEx, UPS). Upgrading shipping tiers will not fix the 80% leakage floor.

Demographics are Irrelevant: Return and cancellation behavior is **uniform across all ages and genders**. The issue is systemic and category-based, not customer-based.

Volume Strain: High-volume categories like **Personal Care** and **Fashion** drive the most leakage. As volume scales, internal quality control currently dilutes.

High-Ticket Stability: Products with **higher unit costs** (e.g., Fitness Equipment) show **higher commitment** and better revenue retention.

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4. Strategic Priorities (Recovery Roadmap)

To recover the **₹2.83M** in lost revenue, the focus must shift to:

Catalog Integrity: Audit high-return categories to ensure **accurate descriptions** and **sizing guides**.

Payment Risk Mitigation: Implement **stricter verification** for identified "Red Zone" payment and carrier combinations.

VIP Protection: Apply **"White Glove" fulfillment** to the **Champion segment** (top 25% of customers), who contribute 50% of the revenue.

Final Recommendations & Business Implications:

1. Reduce High Revenue Leakage Categories

Focus immediately on **categories with high GOV but low NRR**

Perform **product quality audits** for top leakage categories

Improve vendor accountability through **penalty clauses and SLA tracking**

□ *Impact:* Direct reduction in returns and refund losses

2.Optimize Shipping Methods with High Leakage

Identify **shipping methods causing maximum returns and cancellations**

Improve packaging standards for fragile categories

Replace or renegotiate contracts with underperforming logistics partners

 Impact: Lower delivery failures and damage-related returns

3.Strengthen Pre-Delivery Controls to Reduce Cancellations

Improve **delivery promise accuracy** (ETA visibility)

Send proactive order status updates to customers

Introduce confirmation nudges for high-risk orders

 Impact: Reduced cancellations before shipment

4. Improve Product Quality to Reduce Return Rates

Use **return-heavy products** as quality red flags

Enhance product descriptions, images, and sizing information

Introduce stricter seller onboarding and periodic reviews

 Impact: Higher customer satisfaction and fewer post-delivery returns

5.Prioritize High-Risk Revenue Segments

Monitor **high-value orders with poor fulfillment efficiency**

Apply stricter verification or packaging checks for such orders

Introduce insurance or controlled delivery workflows

 Impact: Protects revenue-dense orders from leakage

6. Use Leakage % as a Core Business KPI

Track **Leakage % alongside GOV and NRR** in executive dashboards

Set category-wise acceptable leakage thresholds

Use leakage trend monitoring for early risk detection

 Impact: Data-driven, proactive revenue protection